

Effect of Polyethylene Film Thickness on the Antimicrobial Activity of Embedded Zinc Oxide Nanoparticles

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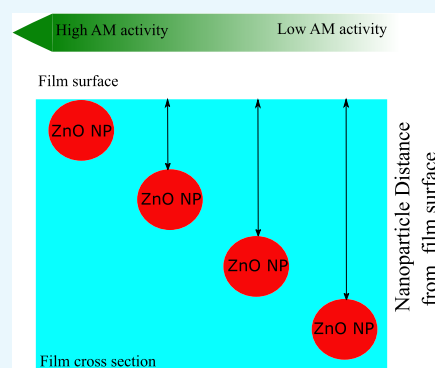


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ABSTRACT: Microbial contamination of most foods occurs primarily at the surface during postprocessing and handling; therefore, preventing cross-contamination by incorporation of antimicrobial substances in contact with the surface of the product is an efficient strategy in reducing food contamination risks. Zinc oxide nanoparticles (ZnONPs) have been used widely to achieve antimicrobial films in various applications including the food industry. This work describes the fabrication of antimicrobial polymeric films containing ZnONPs produced by the coextrusion and dip-coating techniques. Effects of skin layer thicknesses containing ZnONPs on the antimicrobial effectiveness of the film by their capability to inactivate Gram-positive and Gram-negative bacteria were studied for both methods. The antimicrobial properties of the coextruded multilayer LLDPE/ZnONP nanocomposite films evidenced antimicrobial activity in the range 0.5–1.5 log reductions, while in the case of a sandblasted multilayer film, it showed high antimicrobial properties as around 99.99%. The optical properties of coextruded multilayer films were measured and discussed. Furthermore, to achieve a thinner LLDPE thickness, ZnONPs were coated with different concentrations of LLDPE solution by the dip-coating method. TEM confirmed that a homogeneous layer is formed on the surface of ZnONPs. The thickness of the LLDPE layer estimated by TEM was about 2 nm and film produced 3 log and 4 log reductions for *E. coli* and *S. aureus*, respectively. The results show that developed films have the potential to be used as food packaging films and can extend shelf life, maintain quality, and assure the safety of food. The antimicrobial mechanisms of ZnONPs were also investigated. It was found that direct contact of particles with products is necessary to assure high antibacterial activity of the films.



1. INTRODUCTION

Foodborne diseases are occasionally a major cause of death. Thus, protecting the health of the consumer by providing food safety and maintaining food quality is a concern for the food packaging industry. To achieve these goals, the food packaging industry is driven to search for inventive technologies to reduce the risk of pathogen contamination and development in raw and processed foods.^{1–4}

ZnONPs are inorganic, nontoxic, and stable compounds widely used in drug delivery, cosmetics, dentistry, medicines, and food packaging materials.^{5–9} Additionally, ZnO has been considered as a generally recognized as safe (GRAS) material by the US Food and Drug Administration (21 CFR 182.8991).^{10,11} ZnONPs have emerged as appropriate candidates for food packaging due to their excellent antimicrobial activity and can provide a perfect dynamic role for preserving food against food-borne pathogens when they are used in polymeric film packaging.¹²

There are several studies in literature dealing with the influence of the shape,^{13–17} size,^{18–20} concentration,^{19,21} surface modification,^{22,23} and UV illumination²⁴ on the antibacterial activity of ZnONPs. Also, ZnONPs^{25–30} have been widely introduced in different polymeric matrices

including low-density polyethylene (LDPE), high-density polyethylene (HDPE), polypropylene (PP), poly(butylene succinate) (PBS), polyurethane (PU), paper, and chitosan to develop antimicrobial packaging. Previous researches have shown that the antibacterial activities of ZnONPs are greatly reduced after incorporation in polymer matrices. Rojas et al.³¹ successfully prepared low-density polyethylene (LDPE)/ZnONP nanocomposite films by melt compounding and studied the influence of modification and proportion of ZnONPs (3, 5, and 8 wt %) as well as white/UV lights irradiation on the antibacterial and mechanical characteristics of the films. As ZnONP content increased, the antibacterial rate was improved against *E. coli*. At low ZnONP concentration (3 wt %), the surface modification of ZnONPs with oleic acid enhanced the antimicrobial properties, as

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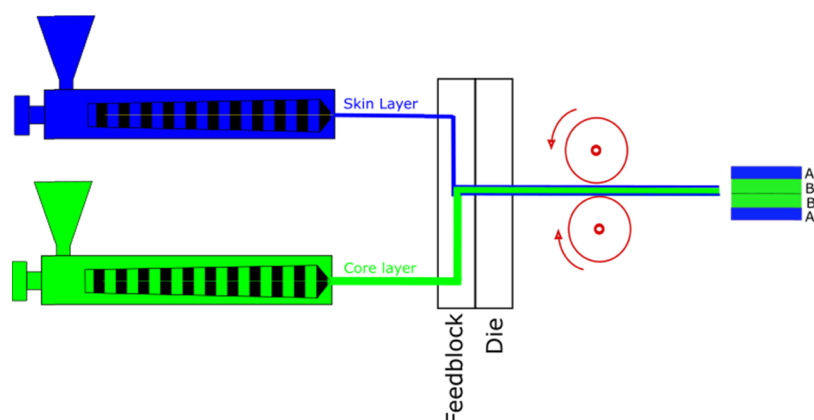


Figure 1. Four-layer film preparation with the coextrusion cast process.

revealed by the bacterial reduction from ~ 33.87 to $\sim 82.26\%$. All of the LDPE/ZnONP nanocomposite films after UV irradiation were effective against *E. coli* with a bacterial reduction of $\sim 99.99\%$ regardless of the nanoparticle concentration and surface modification. In addition, nanocomposites irradiated with white light showed increased effectiveness with nanoparticle incorporation. High antibacterial properties, $\sim 99\%$ against *E. coli*, were achieved for high ZnONP concentration (8 wt %) regardless of the nanoparticle modification.

Li et al.³² tested the antimicrobial activity of nanocomposites of high-density polyethylene (HDPE) and different content of modified ZnONPs (0.2, 0.5, 1, and 2 wt %). The ZnONP surface was modified by γ -aminopropyltriethoxy silane. When the concentration of modified ZnONPs was only 2 wt %, a strong antimicrobial effect of (HDPE)/ZnONP nanocomposite films against *E. coli*, with microbial reduction up to 97.7%, was observed. Petchwattana et al.³³ produced poly(butylene succinate) (PBS) films containing ZnONPs using an extruder and a film blowing unit at five ZnONP contents ranging from 2 to 10 wt %. The results showed that a minimum of 6 wt % ZnONP content was required to inhibit the *E. coli* and *S. aureus* growths with the clear zone of 1.31 and 1.25 cm, respectively.

The antimicrobial properties of ZnONPs³⁴ may be attributed to a direct interaction between ZnONPs and the cell wall,^{20,35,36} release of Zn^{2+} ions,^{36–38} and reactive oxygen species (ROS) formation.^{6,39,40} The theory behind this investigation is that the position of ZnONPs inside the polymeric film plays a vital role in antimicrobial properties. The ZnONP position can be defined by the distance of the particles from the surface of the film. Actually, it can be assumed that particles placed on the top of the film will cause a higher antimicrobial property of the film compared to particles that are completely embedded inside the film. In other words, the possible contact with microorganisms and the release of Zn^{2+} ions into the surrounding medium is definitely stronger, if the particles are directly present close to the film surface.⁴¹ ROS lifetime and potential migration out of the film are expected to be higher when ZnONPs are closer to the film surface.

Thus, in this work, we investigated the effect of LLDPE skin layer film thickness obtained by coextrusion and dip-coating techniques on the antimicrobial activity of incorporated ZnONPs. In the coextrusion method, the produced films are investigated for antimicrobial effects against *E. coli* and *S.*

aureus by LLDPE skin layer film thickness reduction. In the dip-coating technique, different concentrations of LLDPE solution (0.2–0.8 wt %) were used for coating of ZnONPs to find the maximum LLDPE film thickness with the highest antimicrobial activity.

2. MATERIALS AND METHODS

2.1. Materials. ZnONPs (20 nm particle size and the specific surface area of $30\text{--}50\text{ m}^2/\text{g}$) were supplied by SkySpring Nanomaterials Inc. (Houston, TX). Food grade FPs317-A Octene sLLDPE film resin purchased from Nova Chemicals (Calgary, Alberta, Canada) was used as a matrix and coating polymer solution. A low-molecular-weight ethylene-propylene copolymer/acid-modified type (HW 1105A) from MITSUI HI-WAX (molecular weight 1500 and density 939 kg/m^3) was used as a dispersant. Xylene solvent was purchased from Sigma-Aldrich. *E. coli* (DH5 α) and *S. aureus* (ATCC 6538P) were obtained from the Department of Microbiology, Infectiology, and Immunology at the University of Montreal. Luria-Bertani (LB) medium and all of the other chemicals used for growing and maintaining bacteria were purchased from VWR International, LLC.

2.2. Masterbatch Preparation. A masterbatch of LLDPE/ZnONPs was compounded with concentrations of 10 wt %, in the form of pellets using a 18 mm diameter with L/D = 40 twin-screw extruder (Leistritz ZSE 18 HP, Nuremberg, Germany). To keep the final concentration of ZnONPs in the active skin layer at 3 wt %, the masterbatch was diluted during film extrusion. A temperature profile of $195\text{--}220\text{ }^\circ\text{C}$ was used from the melting to die zones in the twin-screw extruder.

2.3. Film Preparation. Multilayer films prepared with a coextrusion cast process had four layers:

- LLDPE/ZnONP layer with concentrations of 3 wt % (diluted masterbatch as a skin on two film sides)
- LLDPE as a core layer.

The film samples were prepared using a Labtech Engineering Coextrusion Multilayer cast film line with two single-screw extruders (type LE20–30 C). The extruders had a 20 mm diameter screw with L/D = 30. The feed-block with (A/B/B/A) design for four layers and a flat die with 30 cm width were used to process the films. The temperature profile for both extruders along the barrel was $200/210/220\text{ }^\circ\text{C}$, and the die and feedblock temperature was $235\text{ }^\circ\text{C}$. The coextrusion line was equipped with a slit die with 2 mm opening. The prepared film was cooled at the die exit using an air knife operating at 13

psi. Chill rolls set at 77 °C were used to stretch the films to obtain different thicknesses in skin layer. Figure 1 shows a schematic view of four-layer films preparation with a coextrusion cast process.

2.4. Preparation of Dip-Coated ZnONPs. In the ZnONP dip-coating process, 0.8, 0.4, and 0.2 g of LLDPE was dissolved in 100 mL of xylene in a conical flask for 60 min at 100 °C. The final suspension was prepared by placing 0.24 g of ZnONPs in all of the LLDPE solutions and stirring for a 24 h to obtain coated ZnONPs. Coated ZnONPs were collected by vacuum filtration and dried thoroughly in a vacuum drier. Table 1 shows ZnONP coating parameters in detail.

Table 1. Composition of the LLDPE Solution for ZnONP Coating^a

sample code	LLDPE concentration (wt/vol xylene %)	mass ratio LLDPE to ZnONPs	estimated coating thickness (nm)
LLDPE 0.8%	0.8	1:2	8 ± 0.2
LLDPE 0.4%	0.4	1:4	4 ± 0.2
LLDPE 0.2%	0.2	1:8	2 ± 0.1

^aThe numbers after ± sign are standard deviations.

2.5. Morphology. The ZnONP distribution in the surface and cross-section of LLDPE/ZnONP films was assessed using a tabletop SEM (TM3030plus, Hitachi) operating at 15 kV acceleration voltage. Skin layer thicknesses were determined using SEM software. The reported layer thicknesses were averaged over 10 different measurements. A transmission electron microscopy (TEM) from JEOL JEM-2100F, Japan, operating at 200 kV was used to examine the surface condition of coated and noncoated ZnONPs.

2.6. Optical Properties. Optical properties of the films in terms of haze values were evaluated in accordance with the procedure specified in ASTM D 1003 using a LAMBDA 1050 spectrophotometer from PerkinElmer.

2.7. Thickness of Films. The film thickness was measured at six random positions using a digital micrometer (ProGage Thickness Tester, Thwing-Albert Instrument Company, NJ), and the mean value was determined.

2.8. Antimicrobial Test. All samples with different skin layer thicknesses were tested against *E. coli* and *S. aureus*. These bacteria were grown in LB broth for 24 h at 37 °C, to reach a density of 10⁸ colony forming units (CFU)/mL. The bacteria culture was diluted in the same medium after 24 h to 10⁵ CFU/mL. Antimicrobial tests were conducted according to the standard ISO 22196:2011 and as described by Abdali and Ajji.⁴² Both active and control samples were cut out into 50 mm × 50 mm coupons. All sample coupons were disinfected by placing under a UV light in a class II biosafety cabinet for 20

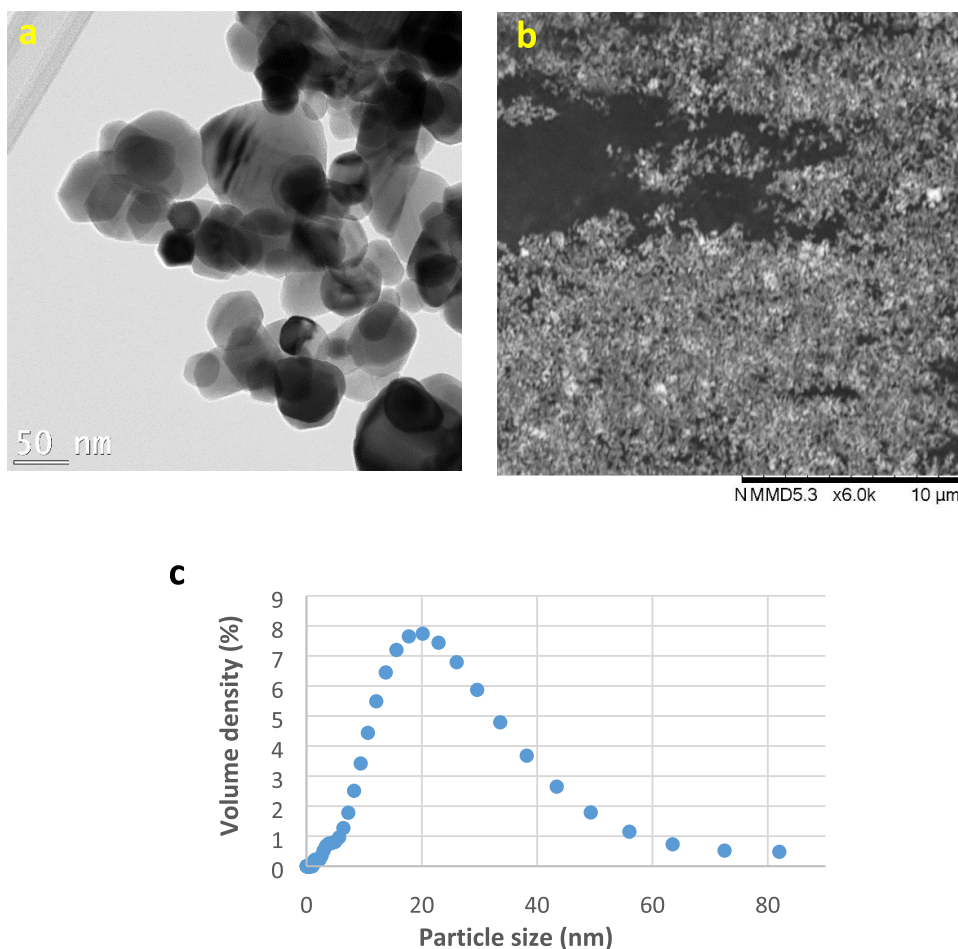


Figure 2. TEM image of ZnONPs (a), SEM of ZnONPs (b), and the particle size distribution of ZnONPs (c).

min. Four hundred microliters of the diluted bacteria culture were deposited on each sample coupon placed in a sterile Petri dish and overlaid with a sterilized plastic thin film [40 mm × 40 mm] for spreading. All Petri dishes containing inoculated sample coupons were incubated at 37 °C for 24 h. The next day, the bacterial inoculums thus placed on the surfaces were washed off with 2 mL of PBS solution. Then, six serial PBS dilutions (10^{-1} to 10^{-6}) were conducted. Three samples were taken from each of the six dilutions, and 10 μ L of droplets were applied onto LB agar plates, which were incubated overnight at 37 °C for counting the surviving bacteria (CFU/mL). Each sample is made in triplicate, along with the negative control.

2.9. Statistical Analysis. Data are the mean $\pm$ SD of three replications. One-way analysis of variance (ANOVA) and Tukey tests were performed to determine significant group differences, and means were considered as statistically significant if $p < 0.05$.

3. RESULT AND DISCUSSION

3.1. ZnONP Characterization. The size effect of ZnONPs in biological interactions and maximizing the generation of reactive oxygen species (ROS) has been thoroughly discussed in literature.⁴³ In other words, the antibacterial activity depends on the concentration and size of ZnONPs.^{19,39,44,45} Figure 2 shows the TEM, SEM, and size distribution of ZnONPs. ZnONPs were spherical in shape and are uniform in size as well as regular in shape as depicted by the TEM image (Figure 2a). The SEM figure (Figure 2b) exhibits a uniform structure and size for ZnONPs. However, in some places, the size of particles is bigger, and they are agglomerated by different groups. The size histogram of ZnONPs in water is shown in Figure 2c. The histogram indicates that the mean particle size of ZnONPs was 20 nm, and the particle size distribution is relatively uniform.

3.2. Thickness. To produce different film thicknesses with different skin layer thicknesses, screw speeds were set at 20, 10, and 5 rpm for the skin layer extruder and 60 rpm for the core layer extruder for all samples. In addition, draw-down ratio (ratio of the roll speed to the extrudate velocity at the die exit) was approximately set at 1.02 (Table 2). The first and last two

Table 2. Sample Codes for Multilayer Films with 3% (w/w) ZnONP Incorporation into the Active Skin Layer^{a,b}

sample code	skin layer extruder screw speed (rpm)	film thickness (μ m)	skin layer thickness (μ m)
M2007	20	103 $\pm$ 2	12 $\pm$ 0.2
M2010	20	70 $\pm$ 3	5 $\pm$ 0.4
M2015	20	89 $\pm$ 1	4 $\pm$ 0.2
M1007	10	82 $\pm$ 3	6 $\pm$ 0.3
M0507	5	117 $\pm$ 1	2 $\pm$ 0.1
M0510	5	66 $\pm$ 2	1.5 $\pm$ 1

^aDraw-down ratio for all samples was set at 1.02 approximately. ^bThe numbers after $\pm$ sign are standard deviations.

digits of each sample code refer to skin layer extruder speed (rpm) and master calendar speed (ft/min), respectively. The thickness of the LLDPE/ZnONP skin layer was determined by SEM analysis. It was found to be in the range of 1.5–12 μ m. The thickness of the whole film specimens and thickness of LLDPE/ZnONP skin layers are shown in Table 2. The mentioned thicknesses were significantly affected by skin layer extruder screw speed.

3.3. Characterization of the LLDPE/ZnONP Film. To check the dispersion of 3% (w/w) ZnONPs into the LLDPE matrix, SEM analysis on the cross-section and surface of LLDPE/ZnONP films was carried out in five random locations around each film sample. SEM of cross-section and surface of LLDPE/ZnONP films are illustrated in Figures 3 and 4, respectively. ZnONPs can be detected on the surfaces of films in the dispersed state. The cross-section of LLDPE/ZnONP films showed nanoparticles were well distributed within the different thicknesses of the skin layer of the multilayer films with no severe aggregations. In general, the SEM images of the film surface of all samples did not show any significant differences. All of the films showed homogeneous and smooth surface structures may be due to the presence of the ethylene-propylene copolymer/acid-modified type. Since this compound have high affinity to metal oxide nanoparticles, it is useful as a dispersing assistant for ZnONPs.⁴⁶ Our study is good agreement with results from Emamifar et al.,²⁶ who concluded that ZnONPs were well dispersed at low concentrations in LDPE films. Similarly, Polat et al.⁴⁷ found that ZnONPs generally displayed good dispersibility in the polypropylene (PP) matrix.

3.4. Optical Properties of LLDPE/ZnONP Films. The presence of ZnONPs reduces the clarity and increases the haze of the films. The LLDPE/ZnONP film samples (M0510 and M2007) showed a little higher haze compared to the neat LLDPE control film with a thickness of 93 μ m due to low ZnONP concentration of 3% (w/w), as illustrated in Figure 5. Similar to our results, Espejo et al.⁴⁸ reported that low-density polyethylene filled with 2% (w/w) ZnONPs was prepared by extrusion and the formed transparent nanocomposite film.

3.5. Antimicrobial Properties of the LLDPE/ZnONP Film. The antibacterial results of all of the samples on the *E. coli* and *S. aureus* are presented in Figure 6. In Figure 6, it is shown that the number of both bacterial colonies dropped with the decreasing of skin layer thickness. It can also be observed that sample M2007 with skin layer thickness 12 μ m present poor antibacterial properties. The thin skin layer films having thicknesses of 4–6 μ m (samples M2015, M2010, and M1007) decreased the *E. coli* growth by less than 1 log (CFU/mL). *S. aureus* was reduced by 1 log (CFU/mL) for samples M2015, M2010, and M1007. Film samples having thicknesses of 1.5 and 2 μ m (samples M0510 and M0507) registered 1 and 1.5 log (CFU/mL) reduction for *E. coli* and *S. aureus*, respectively. These results demonstrated that the antibacterial activity of film samples to inactivate *E. coli* and *S. aureus* was improved by decreasing the skin layer thickness. In other words, a higher thickness of the treated skin layer is a contact barrier between ZnONPs and bacterial cell wall.

To prove that this layer interrupts the antibacterial activity of ZnONPs, a 8 μ m ultrafine sand paper was used to rub the surface of sample M2007. Sanding was performed by hand. The surface was rubbed in a circular motion in two sandblast passages and dust was removed by a vacuum cleaner. This treatment caused increased roughness of the film surface compared to intact sample M2007. This modified film reduced the *E. coli* and *S. aureus* density by approximately 4 log (CFU/mL). The result for the sandblasted film indicated that the antimicrobial activity of the film should be attributed to the direct or very close contact of ZnONPs to the bacteria. The result also agreed with Shen et al. and Rokbani et al. when reported that ZnONPs contact to the cells are required to elicit cytotoxicity.^{49,50}

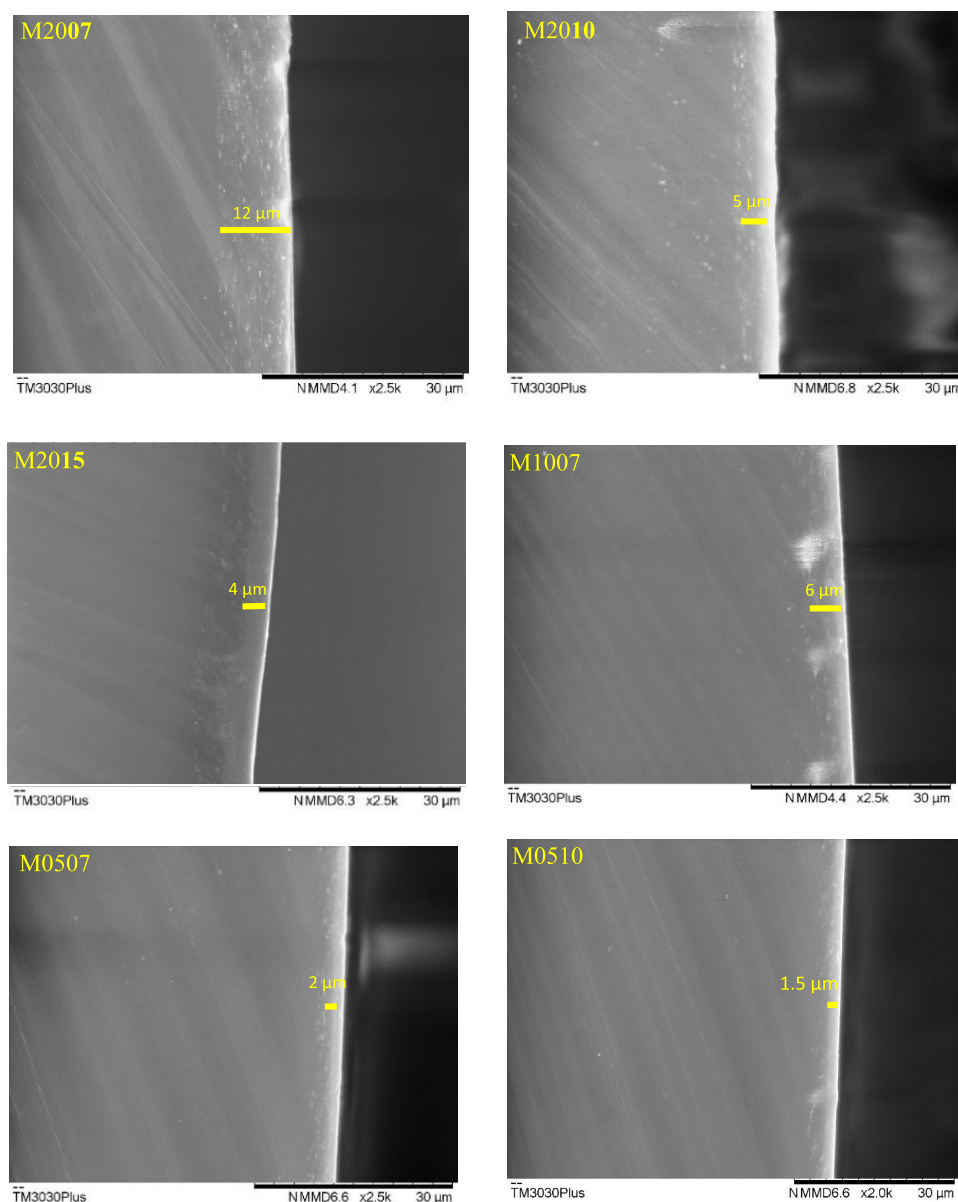


Figure 3. Film cross-section SEM of LLDPE/ZnONPs with 3% (w/w) ZnONP incorporation in the active skin layer.

3.6. Antimicrobial Activity of Dip-Coated ZnONPs. In an attempt to find the maximum thickness of polyethylene between ZnONPs and bacterial cell wall to have proper antimicrobial properties, ZnONPs were dipped into LLDPE solution with different concentrations. The coated ZnONPs were characterized using TEM. The TEM micrographs of coated ZnONPs before and after LLDPE coating can be seen in Figure 7a,b, respectively. The most important feature of Figure 7b is that an irregular-shaped region shows up in middle of Figure 7b, indicating that LLDPE formed a very thin layer thickness (2 nm) in embedded ZnONPs. Figure 8 shows that the thickness of the LLDPE layer on ZnONPs has a large effect on coated ZnONPs. The lower LLDPE concentration decreased the *E. coli* population from 1.6×10^8 to 2.3×10^5 CFU/mL for ZnONPs coated in a solution of LLDPE 0.2%. *E. coli* was reduced more than 1 log (CFU/mL) for both LLDPE 0.4% and LLDPE 0.8% samples. For the Gram-positive strain (*S. aureus*), ZnONPs coated in a solution of LLDPE 0.2, 0.4, and 0.8% decreased the *S. aureus* growth by approximately 4, 3,

and 2 log (CFU/mL), respectively. These results proved that decreasing the coating thickness of ZnONPs to 2 nm improved the antibacterial activity. In general, the results confirm that the antibacterial activity of ZnONPs is a consequence of its direct or very close contact with the microbial cell wall resulting in cell integrity disruption, generation of reactive oxygen species (ROS), and release of Zn^{2+} ions.⁴³ It can also be concluded that the thickness of the coating with different concentrations of LLDPE modulates the antimicrobial activity of ZnONPs. Our results are in good agreement with Saekow et al.,⁵¹ who evaluated the antifungal activity of carboxymethyl cellulose (CMC) coating combined with ZnONPs on persimmon and tomato fruits.

4. CONCLUSIONS

LLDPE/ZnONP skin layer films with thickness varying from 1.5 to 12 μm were produced using the coextrusion technique. To better understand the effect of required LLDPE thickness to achieve a desired antimicrobial activity in the films, dip-

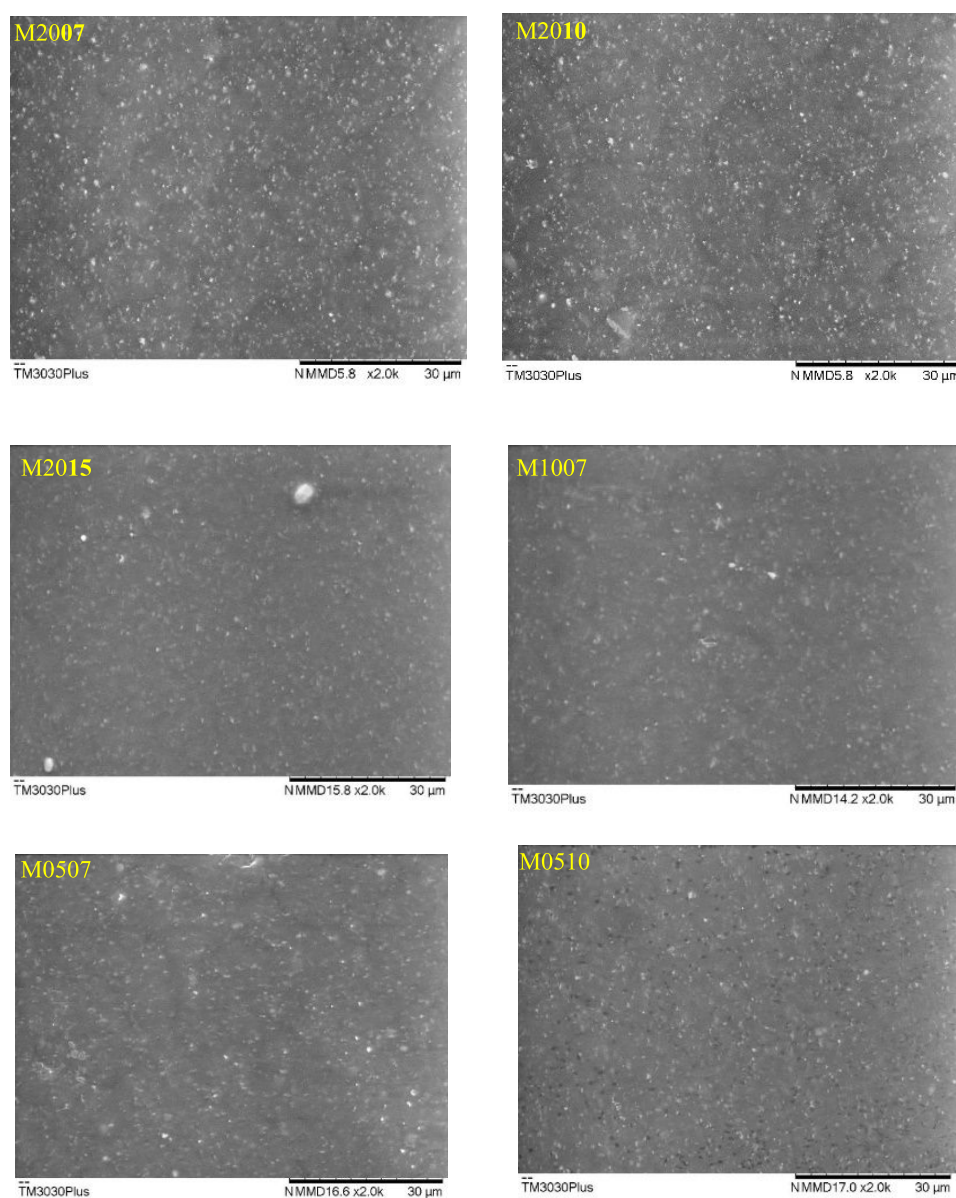


Figure 4. Film surface SEM of LLDPE/ZnONPs with 3% (w/w) ZnONP incorporation in the skin layer.

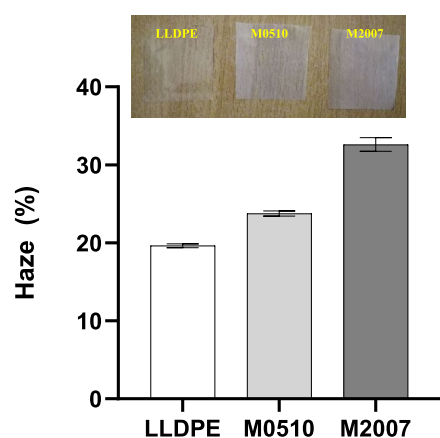


Figure 5. Haze of neat LLDPE and LLDPE/ZnONP film samples (M0510 and M2007) with 3% (w/w) ZnONPs.

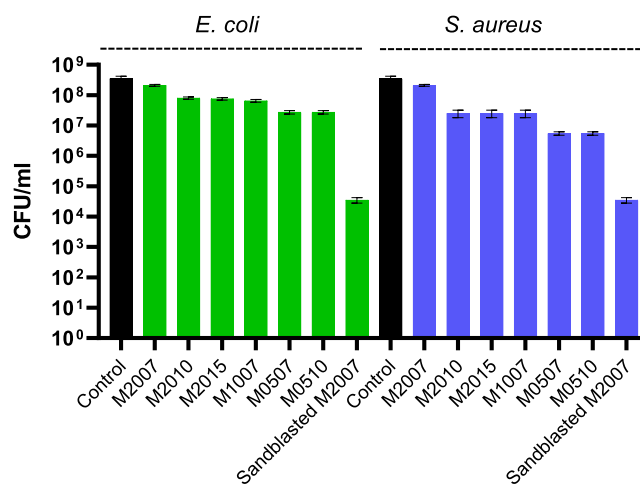


Figure 6. Inhibitory effects of different thicknesses of the LLDPE/ZnONP skin layer film toward *E. coli* and *S. aureus*.

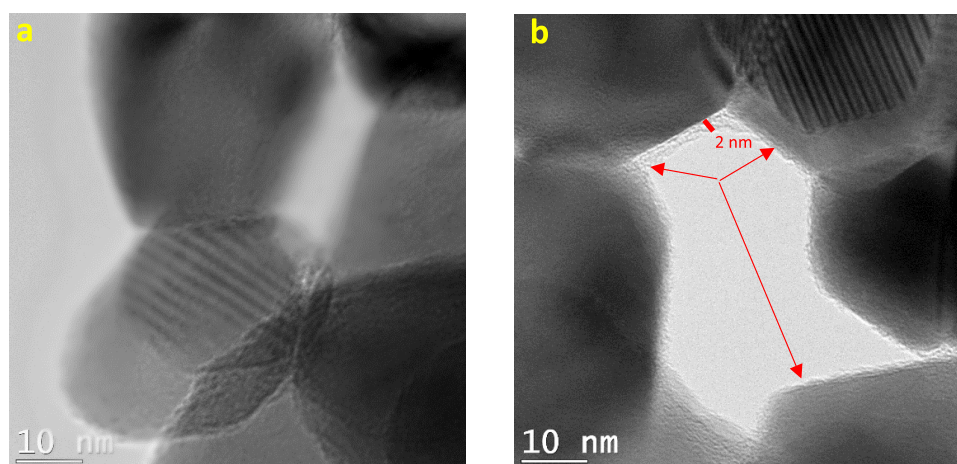


Figure 7. TEM of noncoated (a) and coated (b) ZnONPs with 0.2 wt % LLDPE solution.

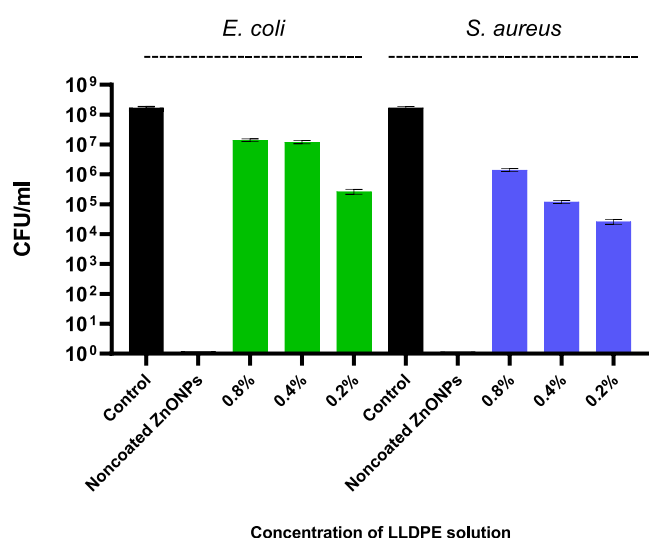


Figure 8. Inhibitory effects of coated ZnONPs in different concentrations of the LLDPE solution toward *E. coli* and *S. aureus*.

coated ZnONPs in different LLDPE solution concentrations and thicknesses closed to 2 nm were also prepared. All of the LLDPE/ZnONP skin layer films and coated ZnONPs showed antibacterial activity. The decrease of LLDPE/ZnONP skin layer thickness down to 1.5–2 μm enhances the antibacterial activity of the film, whereas further decreasing the thickness of LLDPE coating on ZnONPs down to 2 nm dramatically increases the antimicrobial activity of coated ZnONPs. Sandblasting treatment extremely improved multilayer film antimicrobial properties against both *E. coli* and *S. aureus*. All of the films showed homogeneous distribution of ZnONPs. The SEM images of LLDPE/ZnONP films clearly revealed that the ZnONP distribution was uniform. Antibacterial properties of film samples and coated ZnONPs were slightly improved by LLDPE thickness reduction, which may be due to ZnONP direct contact with the microbial cell wall or a better release of reactive oxygen species (ROS), and Zn^{2+} ions from the film surface. Therefore, the proposed LLDPE/ZnONP film with a thin LLDPE inert layer thickness with suitable biocidal properties may be considered as good candidates to be exploited in food packaging.

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Notes

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